

Reg. No.: _____

Amrita Vishwa Vidyapeetham
School of Engineering, Bengaluru Campus
B.Tech. Degree Examination - July 2022
Computer Science and Engineering
Fourth Semester
19CSE213 Operating Systems

Time: Three hours

Maximum marks: 100

CO	Course Outcome
CO1	Understand the architecture and functionalities of modern OS.
CO2	Understand and apply the algorithms for resource management and scheduling
CO3	Apply semaphores and monitors for classical and real world synchronization scenarios
CO4	Understand and apply memory management schemes for modern operating systems

Answer all questions

Differentiate between symmetric and asymmetric multiprocessing systems. [5] [CO1]
Provide a scenario where asymmetric multiprocessing is advantageous. [BTL1]

With the help of state transition diagram explain the various states of a process. [5] [CO1]
[BTL2]

Describe the output of the following C code. [5] [CO1]
[BTL3]

```
#include<stdio.h>
void main(void)
{
    fork();
    fork();
    printf(" Welcome to Amrita\n");
}
```

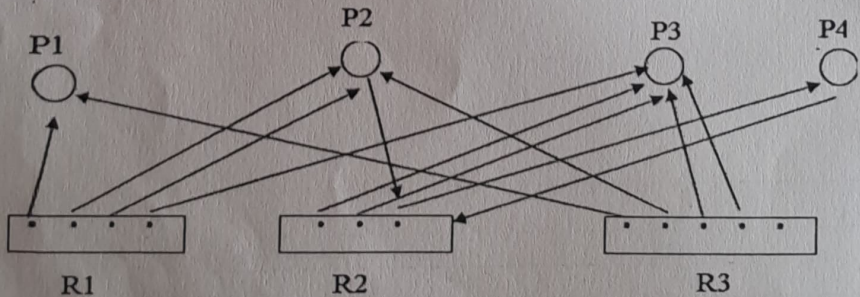
Write a C program to accept 2 strings. Create a child process. The child process [10] [CO2]
should check whether the entered strings are same or not. If they are not same, [BTL6]
check whether the first 4 characters of both strings are same or not. Print suitable
messages. The parent process should wait till the child finishes and then display
a message that "Parent Terminated".

Consider a single-CPU system with only CPU-bound tasks and assume CPU [5] [CO2]
scheduling is always non-preemptive. Also assume each task's runtime is known [BTL4]
in advance, no two tasks have the same runtime, and that all tasks are ready at
system start. Show that Shortest Job First (SJF) scheduling is the sole optimal
scheduling order in terms of mean task turnaround time (where turnaround time
is defined as the elapsed time from when the task enters the system to the
moment the task finishes).

6. Consider the following set of processes with a length of the CPU burst time given in milliseconds.

Process	Arrival Time	Burst Time	Priority
P1	0	5	3
P2	2	2	4
P3	3	4	1
P4	3	1	1
P5	5	3	2

- Draw Gantt charts illustrating the execution of these processes using SRTF, Preemptive priority and round robin (time slice = 1 ms).
 - Compute the waiting times and turn-around time in each of the three schedules and find which of them provides results in the minimal average waiting time and turn-around time.
- Explain the hardware solutions suggested for process synchronization with corresponding code snippets.
 - Explain the solution to Readers-Writers problem using semaphore with proper code snippet.
 - Explain Peterson's solution with a suitable example.
 - Given the resource allocation graph, check whether the system is in a deadlock state. Justify your answer.



11. Consider the following snapshot of a system:

Process	Allocation				Max			
	A	B	C	D	A	B	C	D
P0	1	0	1	0	1	0	2	1
P1	2	1	0	0	2	1	0	2
P2	0	2	0	1	0	3	2	1
P3	1	0	1	0	2	1	1	1
P4	1	2	2	1	3	2	5	2

The available resources are A= 1, B= 1, C= 1, D= 2

- What is the content of NEED matrix?
- Is the system in SAFE state? If so, give the safe sequence.
- If a request from a process P4 (0, 0, 1, 1) arrives, can the request be granted immediately?

Given Memory partitions of 10KB, 45KB, 20KB, 35KB, 48KB (in order), how would each of the Best Fit and First Fit algorithms place processes of 21KB, 39KB, 14KB, and 42KB (in order). Which algorithm makes the best use of memory? [5] [CO4] [BTL2]

Consider a logical address space of 32 pages; each page is 1024 byte long, mapped onto a physical memory of 256 frames. What are the sizes of the logical and physical spaces? Also specify how many bits are there in logical address and physical addresses? [5] [CO4] [BTL2]

On a simple paged system, TLB hold the most active page entries and the full page table is stored in the main memory. If the references satisfied by the TLB take 50 ns, and references through the main memory page table take 400 ns, what is the effective access time if 75% of all memory references find their entries in the TLB? [5] [CO4] [BTL3]

Consider the following page reference string: 1 0 7 1 0 2 1 2 3 0 3 2 4 0 3 0 2 1 0 [10] [CO4] [BTL3]
7. How many page faults would occur for FIFO, LRU and Optimal page replacement algorithms, assuming 5 page frames? Find out which algorithm is suitable for the given scenario.

FOR REFERENCE ONLY

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	15	BTL1	10
CO2	25	BTL2	15
CO3	35	BTL3	47
CO4	25	BTL4	10
		BTL5	8
		BTL6	10

Reg.No. : _____

Amrita Vishwa Vidyapeetham
Amrita School of Engineering, Bengaluru Campus
B.Tech. Degree Examination - January 2020
Computer Science and Engineering
Fourth Semester

15CSE213 Operating Systems

Time: Three hours

Maximum marks: 100

CO	Course Outcomes
CO1	Understand the architecture and functionalities of modern OS and virtual machines
CO2	Understand and apply the algorithms for resource management and scheduling
CO3	Apply semaphores and monitors for classical and real world synchronization scenarios
CO4	Ability to engage in independent learning as a team to study characteristic features of modern operating systems

Answer all questions

Part A

(10 X 3 = 30 Marks)

1. What is solid-state disk (SSD)? List two variants of the same. [CO1]
2. Define System call. Why would an application programmer prefer programming according to an API rather than invoking actual system calls? [CO1]
3. What is program counter? Why is it an important part of Process Control Block? [CO1]
4. Give three programming examples in which multithreading provide better performance than a single-threaded solution. [CO1]
5. Although semaphores provide a convenient and effective mechanism for process synchronization, using them incorrectly can result in errors. List any three scenarios for incorrect use of semaphore operations. [CO3]
6. Mention any three criteria using which various CPU scheduling algorithms can be compared. Write 1-2 lines about each criterion. [CO2]
7. Consider a process which is loaded into memory at address 2001. The number of addresses belong to it is 500. State YES/NO for the following with justification,
 - (i) Can it access the memory location whose address is 1985?
 - (ii) Can it access the memory location whose address is 2175?
 - (iii) Can it access the memory location whose address is 2534?[CO1]
8. Define deadlock. Deadlock can arise if four conditions hold simultaneously in a system. List those four conditions. [CO2]

9. What is wait-for graph in deadlock? Write its purpose.
10. Define the following terms:
 (i) Page fault (ii) Demand Paging (iii) Pager

Part B

(7 x 10 = 70 Marks)

11. (a) Describe the various features of the following design approaches of Operating system with neat diagrams.
 (i) Layered-approach (ii) Micro-kernel approach [6]
 (b) Differentiate between Symmetric and Asymmetric multiprocessing system. [2]
 (c) Define Cache memory. If a cache memory can be made as large as the device for which it is caching (for instance, a cache as large as a main memory), why not make it that large and eliminate the device? [2]
12. (a) Describe scheduler activations with a neat diagram. [6]
 (b) State how an operating system differentiates the parent process from the child process? What are the two execution possibilities that exist for the parent process after creating a child process? [4]

13. Consider the processes listed in Table 1.

Table 1

Process Number	Arrival Time (ms)	Burst time (ms)	Priority
p1	0	7	3
p2	3	2	2
p3	4	3	4
p4	4	15	1
p5	15	3	3

- (i) Draw the Gantt chart to illustrate the execution using Shortest-Remaining-Time-First algorithm (SRTF), Preemptive priority scheduling (smaller number has higher priority) and Round Robin (with time quantum=2). [10]
 (ii) Which of them provide minimal average waiting and turnaround time?
14. (a) Define race condition. List and explain briefly the requirements that a solution to critical section problem must satisfy. [5]
 (b) Write the structure/code of reader and writer processes in the Readers-Writers problem. [5]

15. Consider a system with five processes P0 through P4 and four resource types R1, R2, R3 and R4 as shown in Table 2. Suppose that, at time T0, the following snapshot has been taken:

[CO2]

Table 2

	Max				Allocation				Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P0	0	0	1	2	0	0	1	2	1	5	2	0
P1	1	7	5	0	1	0	0	0				
P2	2	3	5	6	1	3	5	4				
P3	0	6	5	2	0	6	3	2				
P4	0	6	5	6	0	0	1	4				

Using Banker's algorithm

i) Calculate the maximum instances of each resource.

(ii) Compute the need matrix

(iii) Is the system in a safe state? If so, show one sequence of processes which allows the system to complete. If not, explain why. Show your computation step-by-step.

(iv) If a request from process P1 arrives for additional resources of (0, 4, 2, 0), can the request be granted immediately? Show your computation step-by-step.

[10]

16. (a) Given memory partitions of 210KB, 475 KB, 112KB, 300KB, 410KB(in order), how would each of the first fit, best fit, and worst fit algorithms place processes of 212KB, 454KB, 112KB, and 348KB (in order). Which algorithm makes the most efficient use of memory? Show your computation step-by-step.
- (b) Consider a machine with 64MB physical memory and a process of size 8MB. If each page size of the process is 4KB, then answer the following

[CO1]

[6]

[CO1]

- (i) How many pages are there in the process?
- (ii) How many frames are there in the physical memory?
- (iii) How many bits are there in the logical address?
- (iv) How many bits are there in the physical address?

[4]

17. Given references to the following pages by a program

[CO1]

0, 9, 0, 1, 8, 1, 8, 7, 8, 7, 1, 2, 8, 2, 7, 8, 2, 3, 8, 3

How many page faults will occur if the program has initially 3 free page frames available to it and uses the following page replacement algorithms? Show your work out with neat sketch.

- (i) FIFO
- (ii) LRU
- (iii) Optimal

[10]

Reg. No.: _____

Amrita Vishwa Vidyapeetham
School of Engineering, Bengaluru Campus
B.Tech. End semester Examination - May 2025
Electrical and Computer Engineering
Fourth Semester

23ELC211 Operating Systems

Duration: Three hours

Maximum: 100 Marks

CO	Course Outcomes
CO1	Understand the architecture and functionalities of operating systems.
CO2	Analyse process scheduling algorithms.
CO3	Apply the concepts of process synchronization.
CO4	Familiarize storage and memory allocation techniques.

Answer all the questions.

Part-A

- a) Explain the techniques of shared memory and message passing with illustrations. [CO3, BTL2] [5]
- b) Explain Peterson's solutions with code snippets for the classic problem of synchronization. [CO3, BTL2] [5]
- a) Explain the different types of operating systems. [CO1, BTL2] [4]
- b) Explain operating system services in terms of category of services and modules available in the operating system. [CO1, BTL2] [6]
- a) Given 4 processes in Table 1 with burst time and arrival time. Calculate average turnaround time using SJF scheduling algorithm. [CO2, BTL3] [5]

Table 1

Process	Burst time (ms)	Arrival time (ms)
P1	8	0
P2	4	1
P3	9	2
P4	5	3

- b) Given 5 processes in Table 2 with burst time and arrival time. Calculate average turnaround time using priority (pre-emptive) scheduling algorithm (Time quantum = 5s). [CO2, BTL3] [5]

Table 2

Process	Burst time (ms)	Arrival time (ms)	Priority
P1	11	0	2
P2	8	5	0
P3	2	12	3
P4	10	2	1
P5	6	9	4

- 4 a) How do you prevent from the situation of deadlocks? [CO3, BTL2] [6]
- b) Explain safe sequence and safe state conditions. [CO3, BTL2] [4]

Consider the following snapshot of the system

Table 3

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	7	5	3	3	3	2
P1	2	0	0	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	1	2	2	2			
P4	0	0	2	4	3	3			

Answer the following questions using Banker's algorithm:

i) What is the content of the need matrix?

ii) Is the system in a safe state? If yes, mention the safe sequence.

If a request from process P1 arrives for (1,0,2) can the request be granted immediately?

a) Under what circumstances do page fault occur in demand paging environment? Describe the actions taken by the operating system on the occurrence of a page fault

b) What is segmentation? Explain the basic methods of segmentation with example.

a) What is a file? Explain the different attributes of the file system.

b) Discuss the various directory structures with neat diagram.

Given the following disk I/O request sequence:

36, 69, 167, 76, 42, 51, 126, 12. Assume the disk has tracks numbered from 0 to 199, and the disk arm is currently at track 50. What is the total disk travelled by the disk arm using FCFS, SSTF, LOOK and CLOOK algorithm?

Consider the scenario given below:

Set of resources: {R1,R2,R3,R4,R5}

Set of process: {P1,P2,P3,P4,P5}

R1 is allotted to P1 and requested by P2.

R2 is allotted to P2 and requested by P3.

R3 is allotted to P3 and requested by P4.

R4 is allotted to P4 and requested by P1.

i) Draw a resource allocation graph for the above scenario and draw the equivalent wait-for-graph.

ii) Check for the presence of deadlock with justification.

a) Explain contiguous memory allocation in detail

b) Explain how does deadlock and starvation differ from each other

Course Outcomes/ Bloom's Taxonomy Level (BTL) mark distribution table

CO	Marks	BTL	Marks
CO1	10	BTL 1	10
CO2	10	BTL 2	50
CO3	44	BTL 3	30
CO4	36	BTL 4	10
CO5	0	BTL 5	0

[CO3, BTL3] [10]

[CO4, BTL2] [5]

[CO4, BTL2] [5]

[CO4, BTL2] [5]

[CO4, BTL2] [5]

[CO4, BTL3] [10]

[CO3, BTL4] [10]

[CO4, BTL1] [6]

[CO3, BTL1] [4]

Reg.No. : _____

Amrita Vishwa Vidyapeetham
School of Engineering, Bengaluru Campus
B.Tech. Degree Examination –April 2018
Computer Science and Engineering
Fourth Semester
15CSE213–Operating Systems

: Three hours

Maximum marks: 100

Answer all questions
Part A

10 X 3 = 30 Marks

How is availability achieved in clustering? Write any two differences between symmetric clustering and asymmetric clustering.

Parent process at times terminates the execution of one of its child processes. Comment on this statement.

Define signal handling. Mention any two options by which a signal can be delivered in multithreaded programs.

List any three circumstances during which the CPU-scheduling decision is made.

OR REFERENCE ONLY

What are the requirements that need to be satisfied by any solution to the critical section problem? Define them.

Draw a Resource Allocation Graph for the following scenario. There are three process P1, P2 and P3 with four kinds of resources R1 (two instances), R2 (one instance), R3 (three instances) and R4 (two instances). Process P1 is holding an instance of resource R4 and waiting for R2. Process P2 is holding an instance of Resource R3 and waiting for R4. P3 is holding an instance of R1 and waiting for R3. Find whether the system is in Deadlock state or not.

Consider the segment table given in Table 1

Table 1

Segment	Base	Limit
0	220	600
1	2200	14
2	90	100
3	1327	580
4	1952	96

Find the physical addresses corresponding to the logical addresses given below.

i) 0,420 ii) 1,11 iii) 2,500

8. Define the terms: Virtual memory, Physical memory and Demand Paging.

9. When is Direct-access file method used? Give an example.
10. Illustrate the need of a file system? Name the file systems of Windows and Linux.

Part B

7 X 10 =

11. Explain the various features of five design approaches of an Operating System in detail.
12. (a) How does the operating system keep track of various processes which are waiting for shared devices such as magnetic tape and disk drive? Explain your answer with a neat diagram.
(b) State how an operating system differentiates the parent process from the child process? What are the two address-space possibilities that exist for the newly created child process?
13. Consider the processes listed in Table 2. Draw the Gantt chart illustrating the execution using the below mentioned scheduling algorithms. Also, find the average waiting time and turnaround time for each of the scheduling methods.
 - a. Shortest-remaining-time-first algorithm
 - b. Preemptive priority scheduling (smaller number has higher priority)
 - c. Round robin (with time quantum=3)

Table 2

Process Number	Arrival Time (ms)	Burst time (ms)	Priority
p1	0	10	5
p2	0	5	2
p3	2	3	1
p4	5	20	4
p5	10	2	3

14. (a) The basic definition of wait() and signal() semaphore operations suffer from busy waiting. Write a modified definition of wait() and signal() semaphore operations to overcome the busy waiting. Also, explain the modified definitions.
(b) What is Dining-philosopher's problem? Can the Dining-philosopher's problem be solved using semaphores? Justify your answer.
15. Consider a system with five processes P0 through P4 and four resource types R1, R2, R3 and R4 as shown in Table 3. Resource type R1 has 12 instances, R2 has 12 instances, R3 has 8 instances and R4 has 10 instances. Suppose that at time T0, the following snapshot has been taken:

Table 3

	Allocation				Max				Available			
	R1	R2	R3	R4	R1	R2	R3	R4	R1	R2	R3	R4
P0	2	0	0	1	4	2	1	2	3	3	2	1
P1	3	1	2	1	5	2	5	2				
P2	2	1	0	3	2	3	1	6				
P3	1	3	1	2	1	4	2	4				
P4	1	4	3	2	3	6	6	5				

ii. Show that the system is in safe state by demonstrating an order in which the processes may complete.

ii) Suppose now that process P2 requests 2 additional instances of resource type R2, 1 additional instance of resource type R3 and 1 additional instance of resource type R4, Find whether the system is in safe state if this request is granted immediately.

iii) For the same snapshot given in Table 3, Suppose now that process p3 requests 1 additional instance of resource type R1, 1 additional instance of resource type R2, 1 additional instance of resource type R3 and 0 additional instance of resource type R4, can this request be granted immediately. Justify your answer.

16. (a) Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 143, and the previous was at cylinder 125. The queue of pending requests, in FIFO order is: 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for FCFS, SSTF, SCAN scheduling algorithms. [6]
- (b) Compare and contrast on FIFO and Optimal page replacement algorithms. [4]

FOR REFERENCE ONLY

17. (a) List and explain in detail the different ways of hardware implementations of page table. [6]
- (b) Consider a system with logical address space of 512 pages, Size of each page is 4-KBytes and number of frames in physical memory is 128. Answer the following
- i) How many bits are required in the logical address?
- ii) How many bits are required in the physical address? [4]

Reg. No.: _____

Amrita Vishwa Vidyapeetham
 School of Computing, Bengaluru Campus
 B.Tech. Degree Examination – May 2025
 Computer Science and Engineering
 Fourth Semester

23CSE214 Operating Systems

Time: Two hours

Maximum marks: 50

CO	Course Outcomes
CO1	Understand the architecture and functionalities of modern OS
CO2	Understand and apply the algorithms for resource management and scheduling
CO3	Analyze and Apply semaphores and monitors for classical and real-world synchronization scenarios
CO4	Engage in independent learning as a team and implement OS functionalities on a simulated machine.

Answer all questions

Describe how a system ensures that a process cannot access the memory allocated to another process. [5] [CO2, BTL2]

Differentiate between Many-to-one, One-to-one and Many-to-many models of multithreading execution. [3] [CO2, BTL2]

Let there be 5 processes (P1 to P5) and 3 resources (R1, R2 and R3). The maximum number of instances of each resource is 8, 6, and 5 respectively. For each process, the current allocation of resources and their maximum needs are shown in Table 1:

Table 1

Process	Allocation			Max need			Remaining need		
	R1	R2	R3	R1	R2	R3	R1	R2	R3
P1	1	1	1	1	3	2	3	3	0
P2	2	1	1	6	4	4			
P3	1	1	1	5	6	5			
P4	0	0	1	1	0	1			
P5	1	0	1	3	2	3			

- a. Is the above scenario safe? Justify. [3] [CO3, BTL3]
- b. If P3 makes an additional request of (1, 1, 0), should it be granted? Justify. [3]

4 Mention two challenges of multi-core programming using multiple threads, with proper examples. [2] [CO2, BTL2]

5 Consider a scenario where there are 7 Reader processes and 3 Writer processes. Each Reader process attempts to read the k^{th} data within a shared array. The writer processes attempt to write and edit to the same location. Let there be two semaphores: A, a counting semaphore initialized to 5 and a binary semaphore B, initialized to 1. Any number of readers can enter the critical section, but more than one writer cannot enter the critical section at a time. Also, if there is at least one reader, no writer must be allowed in the critical section.

Reader Process

```
while(1){
    wait(B);
    wait(A);
    readcount ++;
    signal(B);
    // Critical section
    wait(B);
    readcount - -;
    signal(A);
    signal(B); }
```

Writer Process

```
while(1){
    wait(B);
    wait(A);
    // Critical section
    signal(A);
    signal(B);
}
```

- a How many Reader processes can enter the critical section at the same time? [1] [CO3, BTL4]
- b Does the above synchronization mechanism guarantee a deadlock safe environment? Justify. [3]

6 Given a six memory partitions of 100 MB, 170 MB, 40 MB, 205 MB, 300 MB, and 185 MB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 200 MB, 15 MB, 185 MB, 75 MB, 175 MB, and 80 MB (in order)? Indicate which all requests cannot be satisfied. Comment on how efficiently each of the algorithms manages memory. [5] [CO2, BTL3]

Consider the following page reference string:

[2+2]

[CO2,
BTL3]

0, 1, 5, 3, 0, 1, 4, 0, 1, 5, 3, 4

Assuming demand paging with three frames, how many page faults would occur for the following replacement algorithms?

i. FIFO replacement ii. Optimal replacement

Compare the page fault rates for the FIFO replacement algorithm with 3 frames, 4 frames, and 5 frames, and analyze how the number of frames affects the page fault rate.

[2+2+
2]

[CO2,
BTL3]

A system is designed where one processor handles I/O and all others handle computation. Is it symmetric multiprocessing or Asymmetric multiprocessing systems? Justify.

[4]

[CO1,
BTL2]

What resources are used when a process is created? How do they differ from those used when a thread is created?

[5]

[CO2,
BTL2]

Demonstrate the concept of virtual machine with a neat diagram.

[6]

[CO1,
BTL1]

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	10	BTL 1	6
CO2	30	BTL 2	19
CO3	10	BTL 3	21
CO4	0	BTL 4	4
		BTL 5	0
		BTL 6	0

23 MAY 2024

Reg. No. : _____

Amrita Vishwa Vidyapeetham

Amrita School of Computing, Bengaluru

B.Tech. Degree Examination – May 2024

Computer Science and Engineering

Fourth Semester

19CSE213 Operating Systems

Maximum marks: 100

Time: Three hours

Course Outcomes
Understand the architecture and functionalities of modern OS
Understand and apply the algorithms for resource management and scheduling
Analyze and Apply semaphores and monitors for classical and real-world synchronization scenarios
Engage in independent learning as a team to study characteristic features of modern operating systems

Answer all questions

- | | | |
|--|-----|--------------|
| 1. Explain the need of an operating system as a <i>control program</i> from a system viewpoint. | 5 | CO1,
BTL2 |
| 2. Draw a diagram illustrating the process of Direct Memory Access (DMA). | 5 | CO1,
BTL2 |
| 3. Discuss the critical role of context switching. | 5 | CO1,
BTL2 |
| 4. Draw a queuing diagram to illustrate the sequence of steps during context switching. | 5 | CO2,
BTL2 |
| 5. You are the system administrator of a multi-user operating system used in a university computer lab. During peak hours, the lab experiences high demand from students working on various tasks, including programming assignments, web browsing, document editing, system updates, etc. | 2+3 | CO2,
BTL3 |
| i) Which CPU scheduling algorithm would be most suitable for this environment?
ii) Justify your choice with necessary diagram. | | |
| 6. i) Draw the state transition diagram of process creation using fork() system call. | 2+3 | CO2,
BTL2 |
| ii) Explain the system calls during the parent-child process relationship. | | |

4. a. i) Predict the output for the following code fragments.

2+2

- ii) Justify the output.

```
int main()
{
    pid_t pid;
    int value=100;
    pid = fork();
    if (pid == 0) /* child process */
        value =200;
    else
    {
        fork();
        printf("The value is %d ", value);
    }
    return 0;
}
```

- b. A set of processes and its corresponding CPU burst time (in milliseconds) is given in the Table 1. 3+2+1

i) Draw the Gantt chart that illustrate the execution of these processes using preemptive priority. (Note: a larger priority number implies a higher priority).

ii) What is the waiting time and turn around time of each process?

iii) Which process has lowest waiting time?

Table 1: Table of processes, and their Arrival time, Burst time, and Priority

Process	Arrival Time	Burst Time	Priority
P ₁	0	2	2
P ₂	0	3	3
P ₃	2	4	5
P ₄	3	6	1
P ₅	5	7	2
P ₆	6	3	4

5. a. Write an algorithm of producer-consumer processes that work in the concurrent computing environment. Give an example to illustrate its application in real world scenario. 5+2

- b. Analyse the potential drawbacks on the algorithm. 3

6. Assume that in a system there are three resources, A, B, and C; and 4 processes P₁ to P₄. The details of the process – resource allocation at time t₀ is given in the Table 2.

Table 2: Details of process – resource allocation

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P ₁	2	1	0	8	6	3	4	3	2
P ₂	1	2	2	9	4	3			
P ₃	0	2	0	5	3	3			
P ₄	3	0	1	4	2	3			

Compute **matrix Need** of Table 2.

2

Is the system in a safe state? If so, show a sequence of processes clearly.

4

Can the request (0,1,0) from P₃ be safely granted for the process-resource allocation in table 2. If granted, will it be safe? Justify.

4

Consider the resource allocation graph given in Fig. 1.

CO3,
BTL3

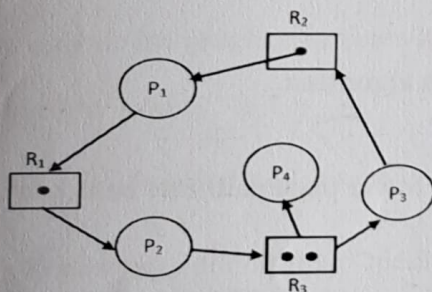


Fig.1. Resource Allocation Graph

Generate its corresponding allocation and request matrix.

2

Identify whether the system generates deadlock or not.

1

Justify your answer by identifying the safe state.

2

Given six memory partitions of 300 KB, 600 KB, 350 KB, 200 KB, 750 KB, and 125 KB (in order), how would the first-fit, best-fit, and worst-fit algorithms place processes of size 115 KB, 500 KB, 358 KB, 200 KB and 275K (in order).

5

CO4,
BTL3

Consider a logical address space of 256 pages with a 4-KB page size, mapped onto a physical memory of 64 frames. [Hint: 1 Kb=2¹⁰ bytes]

CO4,
BTL3

How many bits are required in the logical address?

2½

20 JAN 2023

Reg. No.: _____

Amrita Vishwa Vidyapeetham
 School of Engineering, Bengaluru Campus
 B.Tech. Degree Examination - January 2023
 Computer Science and Engineering
 Fourth Semester

19CSE213 Operating Systems

Time: Three hours

Maximum marks: 100

Course Outcome

Understand the architecture and functionalities of modern OS.
Understand and apply the algorithms for resource management and scheduling
Apply semaphores and monitors for classical and real world synchronization scenarios
Understand and apply memory management schemes for modern operating systems

Answer all questions

How do clustered systems differ from multiprocessor systems? What are the requirements to be met for two machines belonging to a single cluster to offer a highly available service? [5] [CO1] [BTL1]

What is the purpose of the command interpreter? Why is it usually separate from the kernel? Would it be possible for the user to develop a new command interpreter using the system-call interface provided by the operating system? [5] [CO1] [BTL2]

Describe the output of the following C code. [5] [CO1] [BTL3]

```
#include<stdio.h>
void main(void)
```

```
fork();
printf(" Welcome to Amrita\n");
fork();
```

FOR REFERENCE ONLY

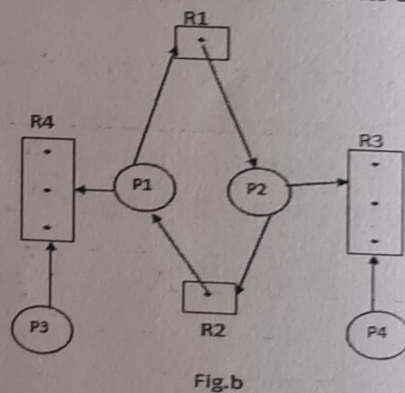
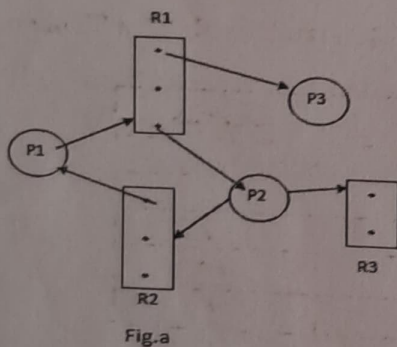
Write a C program to accept a number. Create a child process. The child process should check whether the entered number is palindrome or not. and display appropriate message. The parent process should wait till the child finishes and then display a message that "Parent Terminated". [10] [CO2] [BTL6]

What is a dispatcher? Explain the functionalities of a dispatcher and mention what are the consequences of not having a dispatcher. [5] [CO2] [BTL4]

Consider the following data about process [10] [CO2] [BTL3]

Process	Arrival time	Burst time	Priority
P ₁	0	7	3
P ₂	3	2	2
P ₃	4	3	1
P ₄	4	1	1
P ₅	5	3	3

- i. Draw charts to illustrate execution using SRTF, RR (time slice: 2) and preemptive priority scheduling.
 - ii. Compute waiting time and turnaround time of each process in each of the cases.
 - iii. Compute average waiting time and average turnaround time of each cases.
 - iv. Which of them provide minimal average waiting time and turnaround time?
7. What is Bounded Buffer problem? Discuss briefly with the help of Producer and Consumer Process. [5]
 8. Explain the solution to Dining-Philosophers problem using semaphore with proper code snippet. [7]
 9. Differentiate between starvation and deadlock. Does Peterson's Solution to critical section problem prevents starvation? Justify your answer. [5]
 10. Consider the Resource allocation graph given in Fig. a & b, check whether deadlock exists in the system, if yes identify the process involved in deadlock. [8]



11. Consider the following snapshot of a system [10]

Process	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
P0	0	0	1	0	0	1	1	5	2
P1	1	0	0	1	7	5			
P2	1	3	5	2	3	5			
P3	0	6	3	0	6	5			
P4	0	0	1	0	6	5			

Answer the following questions using Banker's algorithm.

- a) What is the content of the NEED matrix?
 - b) Is the system in a safe state?
 - c) If yes, write the safe sequence.
 - d) If a request from process P1 arrives for (0,4,2), can the request be granted immediately.
12. What do you mean by internal fragmentation and external fragmentation? Provide appropriate example for each. [5]

Consider a logical address space of 8 pages; each page is 2048 byte long, mapped onto a physical memory of 64 frames. What are the sizes of the logical and physical spaces? Also specify how many bits are there in logical address and physical addresses? [5] [CO4] [BTL2]

On a simple paged system, TLB hold the most active page entries and the full page table is stored in the main memory. If the references satisfied by the TLB take 90 ns, and references through the main memory page table take 220 ns, what is the effective access time if 60% of all memory references find their entries in the TLB? [5] [CO4] [BTL3]

Consider the following page reference string: 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6. How many page faults would occur for FIFO, LRU and Optimal page replacement algorithms, assuming 5 page frames? Find out which algorithm is suitable for the given scenario. [10] [CO4] [BTL3]

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	15	BTL1	10
CO2	25	BTL2	15
CO3	35	BTL3	47
CO4	25	BTL4	10
		BTL5	8
		BTL6	10